Documenting a network experiment

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete experiment evidence record.

Keep settings, roles, metric traces, selected state and missing measurements together.

Keep one habit: Recall → W1–W3/W5 → independent record → project manifest → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A useful experiment record explains what was asked, what was done, what happened and what remains unknown.

  Use run NN32-A: does **fitting** one positive ReLU weight reduce validation mean squared error compared with its initial value?

  This tiny constructed regression task teaches record keeping; it cannot establish performance on real-world data.

- The network predicts max(0, a*x), with one weight a and no bias. Start at a = 1, use learning rate 0.1, two full-batch **training** updates and mean squared error, the mean of squared **prediction** errors.

  There is no normalisation, random initialisation or shuffling. Use Python float arithmetic and record the actual Python version when **replay**ing.

- Before running, set the selection rule: choose the **checkpoint** with the smaller validation MSE, breaking a tie toward the earlier update. The initial model is the baseline.

  Validation V1 chooses a **checkpoint** but never contributes to the **training** gradient. Final evaluation has not been run; its result must be recorded as not measured.

Efficiency and resource measurement

> **My remembered idea, example and correction**

Uncertainty and failure analysis

> **My remembered idea, example and correction**

Embeddings and neighbourhoods

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Data role and ID	Input x	Target y
Training T1	1	2
Training T2	2	4
Validation V1	3	6

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Run NN32-A: prediction ReLU(a*x), no bias, initial a = 1, rate .1, two full-batch updates, MSE. Training T1 x1 target 2; T2 x2 target 4. Validation V1 x3 target 6. No preprocessing, random initialisation or shuffle. Positive training preactivations make gradient 5(a–2) valid on this trajectory. Initial training MSE 2.5, validation MSE 9.

  Update 1 gradient –5 gives a1.5, predictions T1/T2 1.5/3 and V1 4.5; training MSE .625, validation 2.25. Update 2 gradient –2.5 gives a1.75, predictions 1.75/3.5 and V1 5.25; training MSE .15625, validation .5625. Minimum validation MSE selects update 2, earlier update wins ties. V1 selected the checkpoint, so it remains validation. Final evaluation, timing, peak memory and energy are not measured.

  Separate picture R8: V4 x2 target 3, weights .5/1/1.25 give predictions 1/2/2.5, squared errors 4/1/.25; choose 1.25. No pictured training trajectory supplied. Keep data manifest, code, settings, all states, raw predictions, metric definitions, selection rule and environment. Hashes detect content changes, not label quality; a repeated replay adds no independent evaluation cases.

- For NN32-A, record the **prediction** rule, initial weight, bias policy, learning rate, update count, loss, **training**/validation IDs, data order or shuffling policy, and **checkpoint** selection rule.

  Name one environment field still to record when you run it.

My reasoning and evidence

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Run NN32-A: prediction ReLU(a*x), no bias, initial a = 1, rate .1, two full-batch updates, MSE. Training T1 x1 target 2; T2 x2 target 4. Validation V1 x3 target 6. No preprocessing, random initialisation or shuffle. Positive training preactivations make gradient 5(a−2) valid on this trajectory. Initial training MSE 2.5, validation MSE 9.

  Update 1 gradient −5 gives a1.5, predictions T1/T2 1.5/3 and V1 4.5; training MSE .625, validation 2.25. Update 2 gradient −2.5 gives a1.75, predictions 1.75/3.5 and V1 5.25; training MSE .15625, validation .5625. Minimum validation MSE selects update 2, earlier update wins ties. V1 selected the checkpoint, so it remains validation. Final evaluation, timing, peak memory and energy are not measured.

  Separate picture R8: V4 x2 target 3, weights .5/1/1.25 give predictions 1/2/2.5, squared errors 4/1/.25; choose 1.25. No pictured training trajectory supplied. Keep data manifest, code, settings, all states, raw predictions, metric definitions, selection rule and environment. Hashes detect content changes, not label quality; a repeated replay adds no independent evaluation cases.

- Calculate both weight updates and give each **training** and validation MSE. Name the selected **checkpoint** and baseline validation MSE. How many independent validation examples support the comparison?

> **My reasoning and evidence**

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Run NN32-A: prediction ReLU(a*x), no bias, initial a = 1, rate .1, two full-batch updates, MSE. Training T1 x1 target 2; T2 x2 target 4. Validation V1 x3 target 6. No preprocessing, random initialisation or shuffle. Positive training preactivations make gradient 5(a–2) valid on this trajectory. Initial training MSE 2.5, validation MSE 9.

  Update 1 gradient –5 gives a1.5, predictions T1/T2 1.5/3 and V1 4.5; training MSE .625, validation 2.25. Update 2 gradient –2.5 gives a1.75, predictions 1.75/3.5 and V1 5.25; training MSE .15625, validation .5625. Minimum validation MSE selects update 2, earlier update wins ties. V1 selected the checkpoint, so it remains validation. Final evaluation, timing, peak memory and energy are not measured.

  Separate picture R8: V4 x2 target 3, weights .5/1/1.25 give predictions 1/2/2.5, squared errors 4/1/.25; choose 1.25. No pictured training trajectory supplied. Keep data manifest, code, settings, all states, raw predictions, metric definitions, selection rule and environment. Hashes detect content changes, not label quality; a repeated replay adds no independent evaluation cases.

- A folder contains only best-model.json and a screenshot of 0.5625. Identify four missing artifacts or records and explain what a content hash can and cannot verify.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Run NN32-A: prediction ReLU(a*x), no bias, initial a = 1, rate .1, two full-batch updates, MSE. Training T1 x1 target 2; T2 x2 target 4. Validation V1 x3 target 6. No preprocessing, random initialisation or shuffle. Positive training preactivations make gradient 5(a–2) valid on this trajectory. Initial training MSE 2.5, validation MSE 9.

  Update 1 gradient –5 gives a1.5, predictions T1/T2 1.5/3 and V1 4.5; training MSE .625, validation 2.25. Update 2 gradient –2.5 gives a1.75, predictions 1.75/3.5 and V1 5.25; training MSE .15625, validation .5625. Minimum validation MSE selects update 2, earlier update wins ties. V1 selected the checkpoint, so it remains validation. Final evaluation, timing, peak memory and energy are not measured.

  Separate picture R8: V4 x2 target 3, weights .5/1/1.25 give predictions 1/2/2.5, squared errors 4/1/.25; choose 1.25. No pictured training trajectory supplied. Keep data manifest, code, settings, all states, raw predictions, metric definitions, selection rule and environment. Hashes detect content changes, not label quality; a repeated replay adds no independent evaluation cases.

- A partner gets different results after changing pre**process**ing and replacing the validation file. They keep the old run ID and overwrite the old score.

  Explain what should be preserved and how the revised work should be recorded.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A separate run R17 records baseline a = 1 and **checkpoint**s a = 1.5 and a = 1.75. It uses **prediction** max(0, a*x) and one validation case V9 with x = 4, y = 8.

  Compute baseline and **checkpoint prediction**s and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Audit a new record independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Add a labelled documenting a network experiment **evidence** sheet to Image-pattern Network Investigation. Show how you can record a network experiment with enough detail to **replay** its calculations.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the selected result for Documenting a network experiment. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Documenting a network experiment. Proposed explanation: "A smaller MSE is an accuracy percentage". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate run R17 records baseline a = 1 and **checkpoint**s a = 1.5 and a = 1.75. It uses **prediction** max(0, a*x) and one validation case V9 with x = 4, y = 8.

  Compute baseline and **checkpoint prediction**s and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Document **checkpoint** selection so another team can reconstruct every reported number.

- Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3. Baseline a = 0.5; **checkpoint**s a = 1 and 1.25; choose minimum validation squared error.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

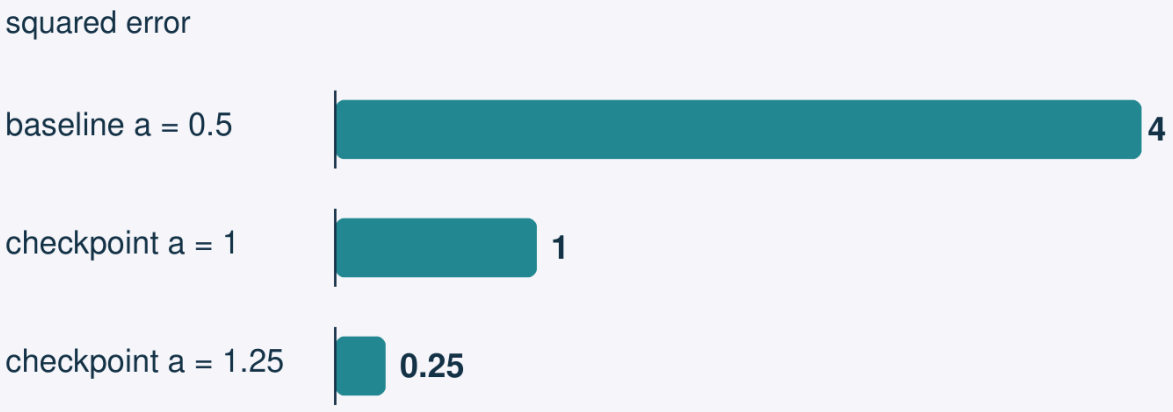
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/32

Validation squared loss by checkpoint



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/32

Documenting a network experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Baseline prediction

At a = 0.5, prediction 1, error2 = 4.



2. Checkpoint comparison

At a = 1, prediction 2, loss 1; at 1.25, prediction 2.5, loss 0.25.



3. Select and record

Choose a = 1.25; save R8, inputs, formula, checkpoint values, loss rule and validation role.

CHANGE ONE THING

A teammate calls V4 a final test after using it to select a=1.25.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Document **checkpoint** selection so another team can reconstruct every reported number.
- Which labelled step matches this detail? Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.
- 1. **Checkpoint** comparison
- 2. Select and record
- 3. Baseline **prediction**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Select and record Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.

A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

Changed-case question and optional hint

- Changed-case practice: A teammate calls V4 a final test after using it to select a=1.25. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Correct its role to validation; it was used for selection.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Correct its role to validation; it was used for selection. Changing the label on a dataset does not make its **evidence** untouched.

Repair and retain the record

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10 ms/case not necessarily a single-request latency?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Documenting a network experiment: the case

AI ACADEMY / AI-5/32 · 01 · LOOK AT THE FACTS

Documenting a network experiment: the case

The situation

Document checkpoint selection so another team can reconstruct every reported number.

Supplied evidence

Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3. Baseline a = 0.5; checkpoints a = 1 and 1.25; choose minimum validation squared error.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain documenting a network experiment using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

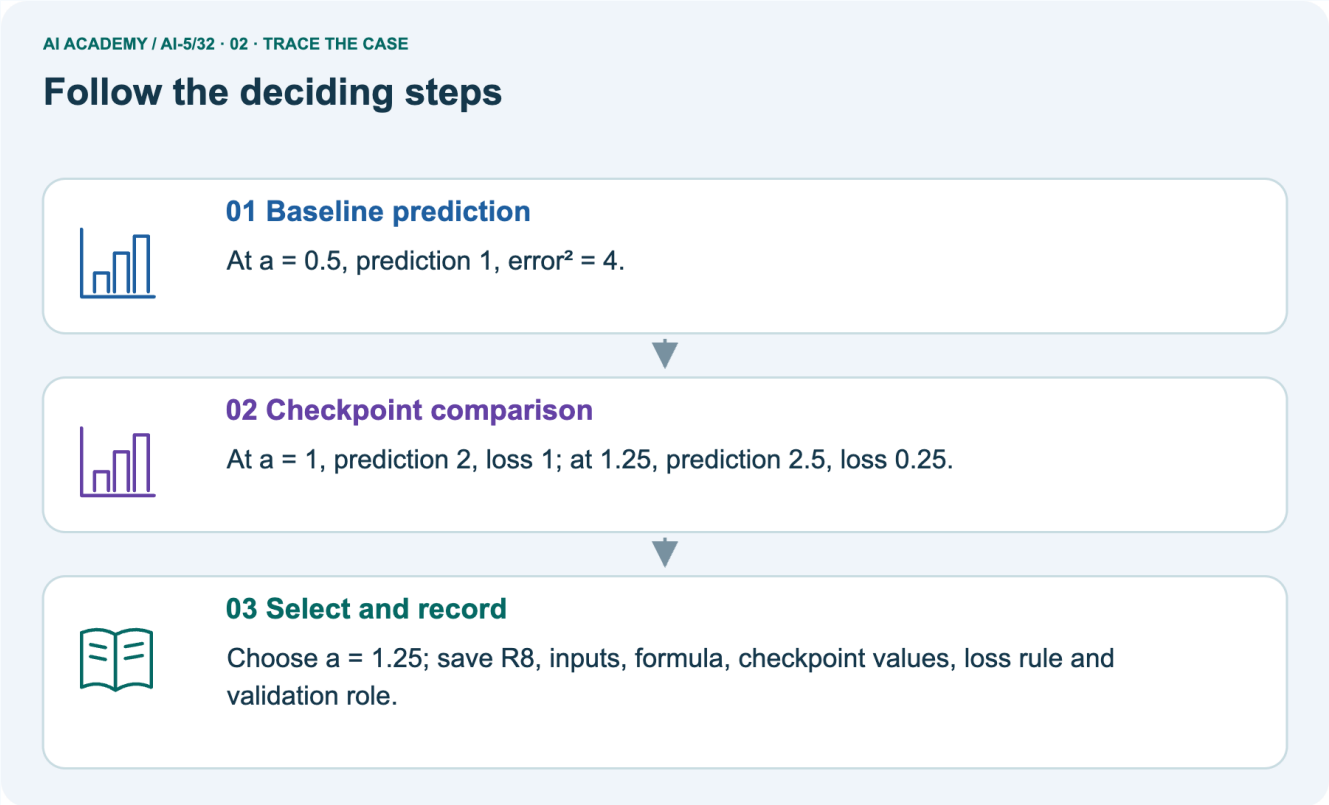
Document **checkpoint** selection so another team can reconstruct every reported number. Supplied facts: Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3.

Baseline a = 0.5; **checkpoint**s a = 1 and 1.25; choose minimum validation squared error. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



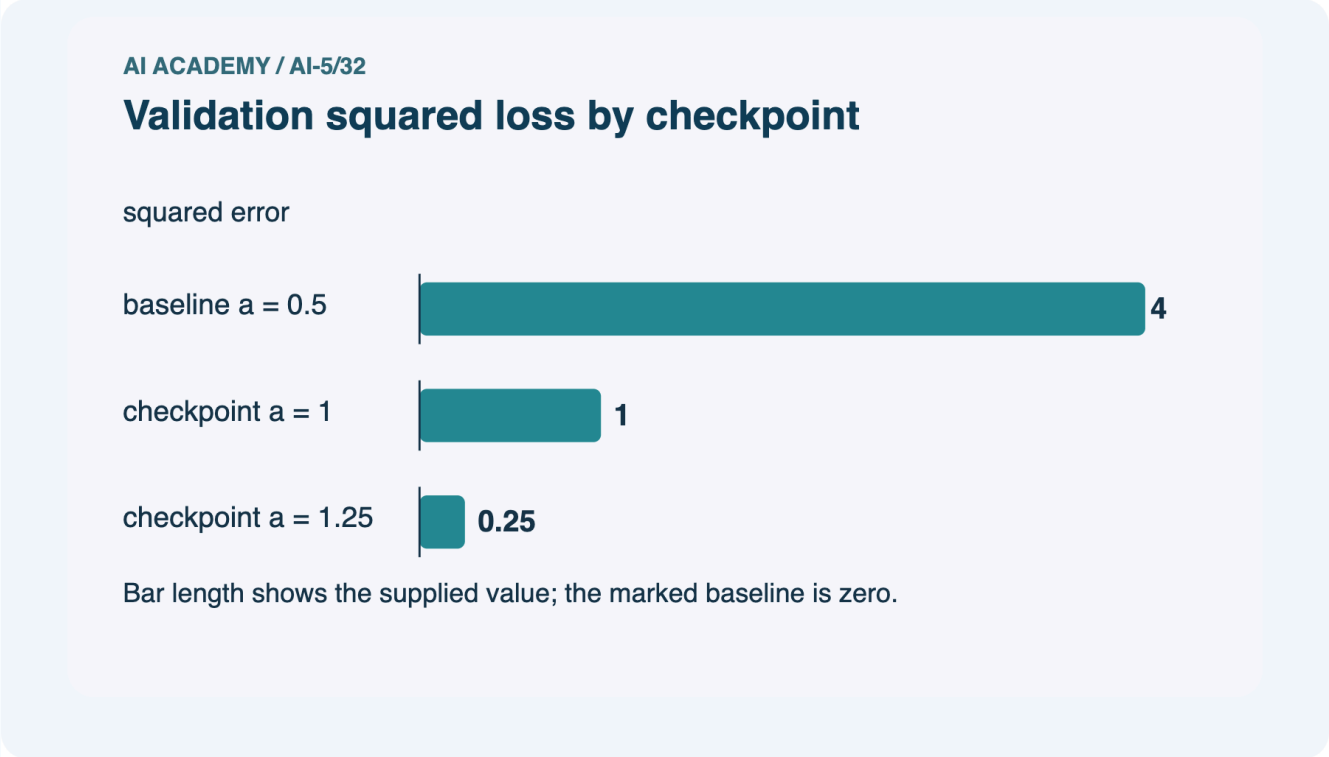
Follow the labelled picture with the matching explanation. Open the vector diagram

Baseline **prediction**: At a = 0.5, **prediction** 1, error² = 4. → **Checkpoint** comparison: At a = 1, **prediction** 2, loss 1; at 1.25, **prediction** 2.5, loss 0.25. → Select and record: Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Validation squared loss by checkpoint



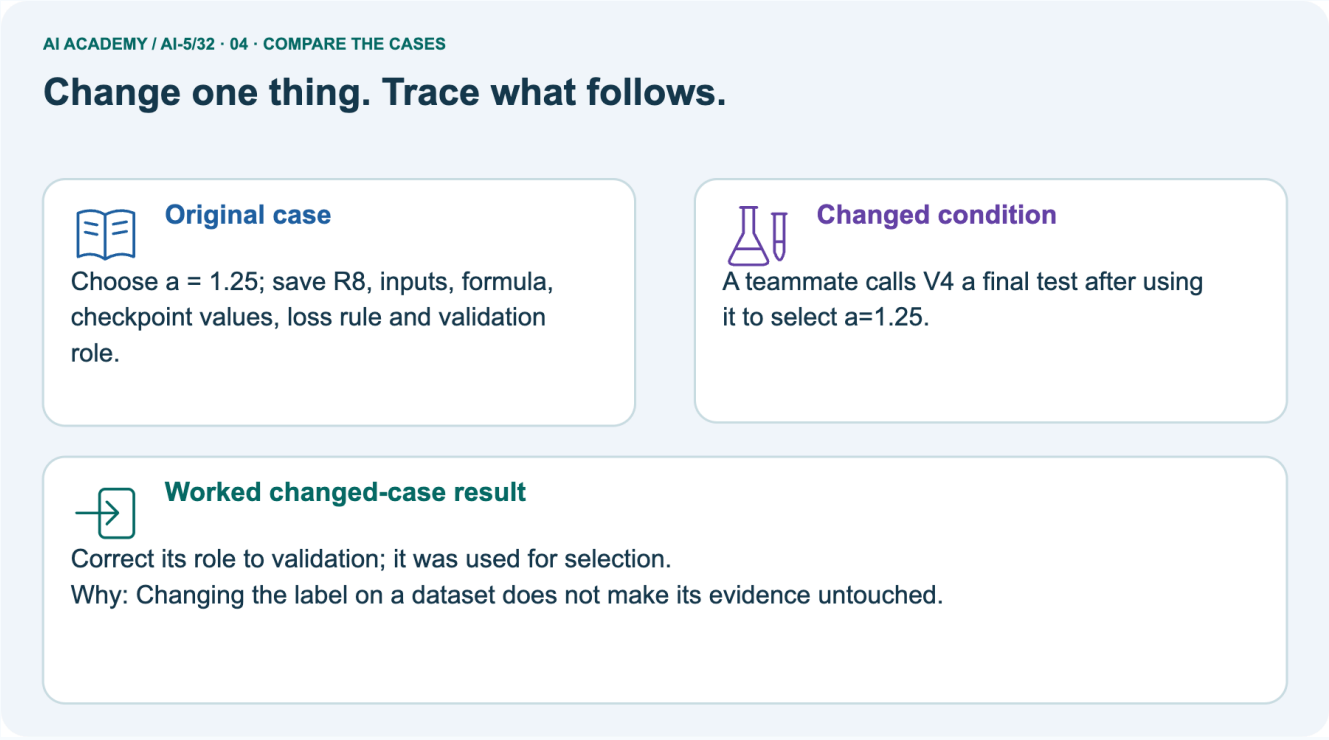
Follow the labelled picture with the matching explanation. Open the vector diagram

Validation squared loss by **checkpoint**. Units: squared error. baseline a = 0.5: 4. **checkpoint** a = 1: 1. **checkpoint** a = 1.25: 0.25.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

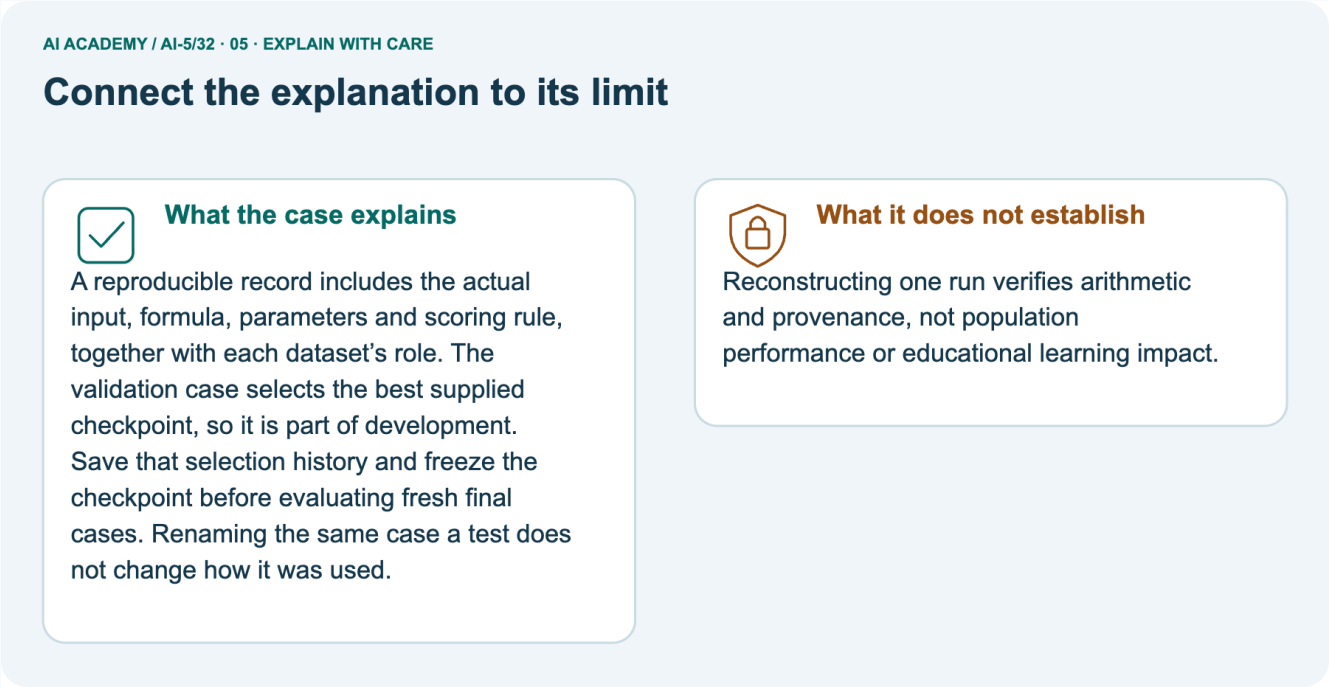
Original case: Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role. Changed condition: A teammate calls V4 a final test after using it to select a=1.25.

Expected result: Correct its role to validation; it was used for selection. Reason: Changing the label on a dataset does not make its **evidence** untouched.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

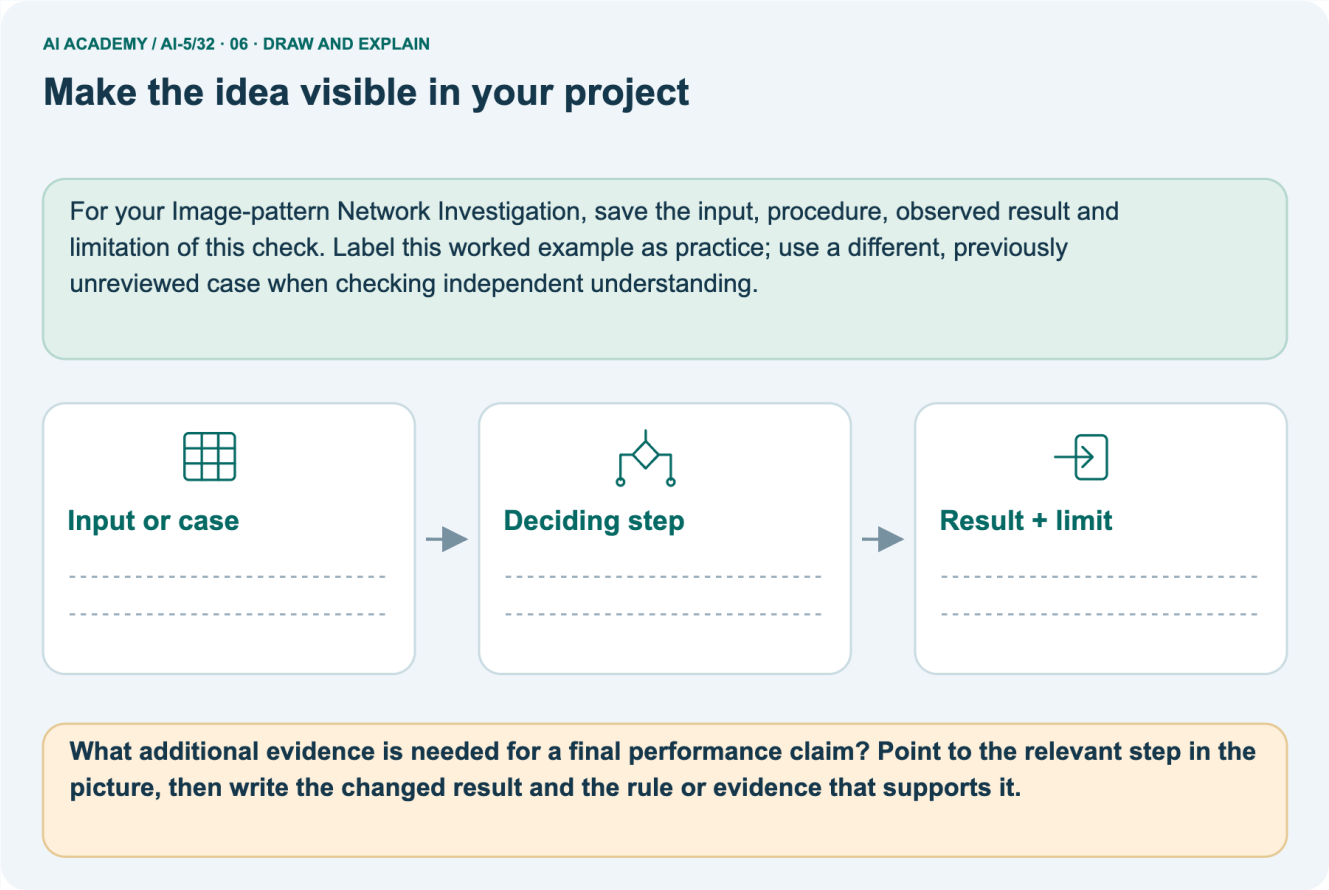
Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

Limit: Reconstructing one run verifies arithmetic and provenance, not population performance or educational learning impact.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Documenting a network experiment: the case

02. Follow the deciding steps

03. Validation squared loss by checkpoint

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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